# Experiment No : 6

**Title :Implement Stack using Linked List**

**Name of Student:Shivam Sevewar**
**Class: E&TC Div: SY1 PRN No:B25ET1042 Batch:B**

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## Program:

```
#include <stdio.h>
#include <stdlib.h>

struct node {
   int data;
   struct node *next;
};

struct node *top = NULL;

void push(int x) {
   struct node *newNode = malloc(sizeof(struct node));
   if (newNode == NULL) {
      printf("Memory not available\n");
      return;
   }
   newNode->data = x;
   newNode->next = top;
   top = newNode;
}

void pop() {
   if (top == NULL)
      printf("Stack is Empty\n");
   else {
      struct node *temp = top;
      printf("Popped: %d\n", top->data);
```

```
        top = top->next;
        free(temp);
    }
}

void display() {
    struct node *temp = top;
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

int main() {
    push(10);
    push(20);
    push(30);

    printf("Stack: ");
    display();

    pop();

    printf("Stack after pop: ");
    display();

    return 0;
}
```

**Output:**

